

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I V.Kailash(192521232)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

v.kailash

APPLICATION- BASED QUESTIONS

1. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Ans: 1. **Objective and Concepts**

Task: Recover the plaintext of the intercepted ciphertext **GSRH RH Z HVXIVG** and justify the result.

Concepts involved

- **Substitution cipher:** each plaintext letter is always replaced by the same ciphertext letter. Word lengths and spaces are preserved.
- **Monoalphabetic vs. polyalphabetic:** one fixed mapping (monoalphabetic) is much weaker than a mapping that changes per position (e.g., Vigenère).
- **Atbash cipher:** a fixed monoalphabetic substitution that reverses the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, and so on).
- **Cryptanalysis techniques:** pattern-word analysis, frequency analysis, crib guessing, and brute-force elimination.

The problem says the structure suggests "a simple substitution mechanism rather than complex encryption." This is consistent with the spaces being preserved and the message being short and readable-looking.

2. Experimental Design (Method)

I used a systematic, ordered approach, moving from observation to hypothesis to verification.

Step	Action	Purpose
1	Observe structure	Word lengths, spaces, single letters
2	Pattern-word analysis	Guess candidate English words
3	Frequency analysis	Check letter counts against English
4	Form hypotheses	Caesar/ROT13 vs. Atbash vs. general substitution
5	Eliminate wrong hypotheses	Test and reject Caesar and ROT13
6	Apply the Atbash mapping	Decode by hand
7	Verify with code	Confirm programmatically
8	Validate the	Check that the plaintext is

	result	meaningful English
--	--------	--------------------

3. Analysis of the Ciphertext

3.1 Structure

GSRH RH Z HVXIVG

4 2 1 6 ← word lengths

Four words, lengths 4-2-1-6, spaces preserved. That strongly indicates a **letter-for-letter substitution** (a transposition would likely scramble word boundaries, and a block cipher would not preserve them).

3.2 Single-letter word

In English, the only one-letter words are **A** and **I**. So **Z = A or I**. "A" is far more common in running text, and it fits a sentence like "X is A Y".

3.3 Two-letter word

RH is a two-letter word. Common English two-letter words are *IS, IT, IN, IF, AT, AS, OF, ON, OR, TO, BE, ...*

The pattern is "two different letters, ending in H", and H is also the last letter of the first word GSRH. So the hypothesis **RH = "IS"** is strong, because "...IS IS A ..." and "THIS IS A ..." are very common constructions.

3.4 Four-letter word

GSRH ends in the same two letters as **RH**. If RH = IS, then GSRH = "??IS".

The word **THIS** fits perfectly (T-H-I-S), giving G = T, S = H, R = I, H = S.

3.5 Six-letter word with repeated letter

HVXIVG:

Position: 1 2 3 4 5 6

Letter: H V X I V G

V repeats at positions 2 and 5, giving the pattern **a b c d b e**. Using the letters already deduced (H = S, G = T), the word is **S ? ? ? ? T** with the second and fifth letters equal. **SECRET** matches (S-E-C-R-E-T), with E repeated at positions 2 and 5.

3.6 Frequency analysis

Counting letters in the ciphertext (13 letters total):

L e t t e r	H	G	R	V	S	Z	X	I
C o u n t	3	2	2	2	1	1	1	1

The most frequent letter, **H**, maps to **S**. The next group (G, R, V) maps to T, I, E, all very common English letters. The sample is tiny, so frequency analysis

is only supporting evidence here, not proof, but it is consistent.

4. Hypothesis Testing

Hypothesis A: Caesar shift (including ROT13)

A Caesar cipher shifts every letter by a constant. From the single-letter word:

- If $Z \rightarrow A$, the shift is +1, so $GSRH \rightarrow HTSI$ (nonsense).
- If $Z \rightarrow I$, the shift is -17 (or +9), so $GSRH \rightarrow PBAQ$ (nonsense).
- ROT13 gives $GSRH \rightarrow TFEU$ (nonsense).

Rejected.

Hypothesis B: Atbash (reversed alphabet)

Atbash needs no key and is a classic "simple substitution." Let us test it.

Rule: if $A = 1$, $B = 2$, ..., $Z = 26$, then plain position = $27 - \text{cipher position}$.

Cipher	Position	$27 - \text{pos}$	Plain
G	7	20	T
S	19	8	H
R	18	9	I
H	8	19	S
Z	26	1	A
V	22	5	E
X	24	3	C
I	9	18	R

These are exactly the same letter assignments that the pattern-word analysis suggested independently ($G = T$, $S = H$, $R = I$, $H = S$, $Z = A$, $V = E$), which is strong confirmation. Two independent routes (pattern words, and one global mathematical rule) agree.

Accepted.

Full Atbash table (reference)

Plain : A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Cipher: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

5. Decoding

Ciphertext : G S R H R H Z H V X I V G

Plaintext : T H I S I S A S E C R E T

Plaintext: "THIS IS A SECRET"

Output

THIS IS A SECRET

True

The round-trip check ($\text{atbash}(\text{atbash}(x)) == x$) confirms the implementation and demonstrates that Atbash is an **involution**: encryption and decryption are the same operation.

A general-purpose alternative is a quick brute-force over all 26 Caesar shifts, which would have shown none of them produce a sentence, pushing the analyst toward a non-shift substitution like Atbash.

6. Validation of Results

Check	Result
Output is grammatical English	Yes: "This is a secret"

Every word is a valid dictionary word	Yes
Word lengths preserved (4-2-1-6)	Yes
Repeated cipher letters ↔ repeated plain letters	Yes (V↔E twice, R↔I twice, H↔S three times, G↔T twice)
Mapping is one-to-one (no collisions)	Yes (8 distinct cipher letters ↔ 8 distinct plain letters)
Mapping follows one global formula (27 – position)	Yes, for all 8 letters
Wrong hypotheses (Caesar, ROT13) produce gibberish	Yes

The chance that a wrong key would produce a fully meaningful English sentence of 13 letters is negligible, so the result is highly reliable.

7. Technical Characterization of the Cipher

- **Type:** monoalphabetic substitution, specifically **Atbash**.
- **Mathematical form:** an affine cipher $E(x) = (a \cdot x + b) \bmod 26$ with $a = 25 (\equiv -1)$ and $b = 25$, so $E(x) = 25 - x$ (using $A = 0 \dots Z = 25$).
- **Key space:** effectively **1** (there is no key), so it offers no secrecy beyond obscurity.
- **Weaknesses:**
 - Fixed mapping, so it is vulnerable to frequency analysis on longer text.
 - Preserved word boundaries expose pattern words (like "A", "IS", "THIS").
 - Known-algorithm attack: just try Atbash.
- **Historical context:** originally a Hebrew cipher (aleph↔tav), used in biblical-era texts.

8. Security Interpretation (for the intern's report)

- The message content ("THIS IS A SECRET") is benign, but it indicates **low-sophistication obfuscation** rather than real encryption.
- Recommended actions: log the intercept, note the source and destination, check for repeated or longer Atbash traffic, and escalate if the same sender uses more complex methods.
- Real confidentiality would require a modern cipher such as **AES** with proper key management, not substitution.

9. Conclusion

The ciphertext **GSRH RH Z HVXIVG** is an **Atbash-encoded** message, and the decoded plaintext is:

"THIS IS A SECRET"

The conclusion was reached by combining pattern-word analysis, frequency observations, elimination of Caesar/ROT13, a single global mapping rule, and programmatic verification.

2. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Ans: 1. Concepts

- Transposition cipher:** letters are rearranged, not replaced. The ciphertext contains exactly the same letters as the plaintext, only in a different order.
- Rail Fence cipher:** the plaintext is written in a zigzag across n rows ("rails"), then read off row by row. "Three levels of arrangement" means **3 rails**.
- Cryptanalysis tools:** letter-frequency check, structural analysis, trying every possible key, and a crib (a guessed plaintext word).

2.

Step 1: Observe the ciphertext

WECRLTEERDSOEFEAOCAVDEN → 25 letters, no spaces

Letter counts: E = 7, then C, D, A, O = 2 or 3 each, with W, R, L, T, S, F, I, V, N appearing once or twice. This is the normal frequency profile of English, with E on top. A substitution cipher would hide that profile, so the letters have only been moved. That points to a transposition cipher, and "three levels" suggests the Rail Fence cipher with 3 rails.

Step 2: Work out the rail lengths

With 25 letters on a 3-rail zigzag, the rail pattern repeats every 4 positions (0-1-2-1-0-1-2-1...):

Rail	Positions used (0-indexed)	Letters
Rail 1 (top)	0, 4, 8, 12, 16, 20, 24	7
Rail 2 (middle)	1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23	12
Rail 3 (bottom)	2, 6, 10, 14, 18, 22	6

Check: $7 + 12 + 6 = 25$.

Step 3: Split the ciphertext into rails

The ciphertext is Rail 1, then Rail 2, then Rail 3, concatenated:

Rail 1 (7 letters) : W E C R L T E

Rail 2 (12 letters): E R D S O E E F E A O C

Rail 3 (6 letters) : A I V D E N

Step 4: Read the zigzag to recover the plaintext

Reading down and up the rails in the pattern 1-2-3-2-1-2-3-2...:

Rail 1: W . . . E . . . C . . . R . . . L . . . T . . . E

Rail 2: . E . R . D . S . O . E . E . F . E . A . O . C .

Rail 3: . . A . . . I . . . V . . . D . . . E . . . N . .

Reading column by column gives:

WEAREDISCOVEREDFLEEATONCE

Step 5: Segment into words

"WE ARE DISCOVERED FLEE AT ONCE"

This is a well-known test phrase for the Rail Fence cipher, and it makes sense as a message.

Step 6: Verify by re-encrypting

Rail	Letters from plaintext	Result
1	W, E, C, R, L, T, E	WECRLTE
2	E, R, D, S, O, E, E, F, E, A, O, C	ERDSOEFEAOC
3	A, I, V, D, E, N	AIVDEN

Concatenated, this gives **WECRLTEERDSOEFEAOCAIVDEN**, which matches the intercepted ciphertext exactly.

3.

	Weakness	Explanation
1	Tiny key space	The only key is the number of rails. For a 25-letter message there are about 23 possible keys (2 to 24 rails). A computer tries them all in microseconds, and a human can do it in minutes (the loop in Step 7).
2	No diffusion	Each plaintext letter is moved to one fixed place. Changing one plaintext letter changes only one ciphertext letter. Modern ciphers (AES) spread every input bit across the whole block.
3	No confusion	The letters themselves are unchanged. The ciphertext has the same letter frequencies as English, so the E-heavy profile reveals immediately that it is a transposition.
4	Structure leaks through	The rail pattern is deterministic. Once a short word is guessed (a crib like "THE", "ATTACK", or "FLEE"), it fixes the key and exposes the rest.
5	Kerckhoffs's principle violated	Security depends on the method staying secret. Rail Fence is a published textbook cipher, so once an attacker knows the method, nothing secret remains.
6	Vulnerable to known-plaintext attack	A single known plaintext/ciphertext pair reveals the number of rails.
7	No integrity or authentication	An attacker can swap or reorder ciphertext blocks, or inject text. The receiver has no way to detect tampering.
8	No semantic security	The same message always produces the same ciphertext, so repeated or predictable messages are recognizable. There is no IV or nonce.
9	Brute-force and automated tools	Automated solvers (frequency scoring, n-gram or dictionary tests) crack transposition ciphers in well under a second, with no knowledge of the plaintext.
10	Not scalable to a corporate setting	There is no key management, no key rotation, no forward secrecy, and no protection against insider or network-level attackers. It fails every modern confidentiality, integrity, and compliance requirement (for example under GDPR, ISO 27001, or PCI DSS).

Overall assessment: Rail Fence is a good teaching tool and historically interesting. It provides **obscurity**, **not security**, and it is unsuitable for any real corporate communication.

Element	Implementation
Cipher	AES-256-GCM , an authenticated encryption (AEAD) scheme that gives confidentiality and integrity together.
Transport	TLS 1.3 for message transport, with forward secrecy through ephemeral Diffie-Hellman key exchange.
Key management	Generate random 256-bit keys, store them in an HSM or key vault (such as Azure Key Vault or AWS KMS), and rotate them on a schedule.
Nonce/IV	A unique random 96-bit nonce for every message, which stops identical plaintexts from producing identical ciphertexts.
Migration path	Add a wrapper or gateway in front of the legacy system so the old software need not be rewritten. The gateway encrypts and decrypts, and legacy endpoints speak to it locally.

6. Final Answers

- **(a)** Original message: **"WE ARE DISCOVERED FLEE AT ONCE"** (Rail Fence, 3 rails).
- **(b)** The method is vulnerable because of its tiny key space, lack of diffusion and confusion, frequency leakage, dependence on a secret method, no integrity protection, and easy automated attacks.
- **(c)** Enhancement: adopt AES-256-GCM over TLS 1.3 with managed keys and unique nonces, deployed through a gateway so the legacy system can migrate without being rewritten.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
---------------------	----	--	--	--------------------------------------	---------------------------------------

Documentat ion	10	Clear, well- structured documentation with diagrams, code, and results	Organized documentati on with necessary details	Basic documentatio n with missing details	Poor or incomplete documentatio n
-------------------	----	--	---	---	--

